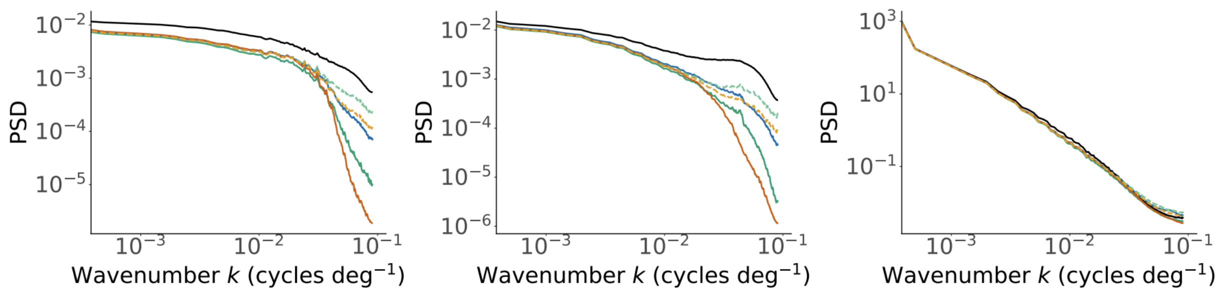
Physical Consistency

Spatial

1. Check that PSD (variance/energy) over wavenumbers between ERA5 and model agree
    a. Mean horizontal kinetic energy
        i. $E_h = \frac{1}{2}(u^2 + v^2)$
    b. Horizontal continuity/divergence:
        i. $\delta_h = \nabla_h \cdot v_h = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$
    c. Relative horizontal vorticity:
        i. $\zeta_h = \nabla_h \times v_h = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$



(d) PSD of $\boldsymbol{\delta_h}$ (e) PSD of $\boldsymbol{\zeta}_h$ (f) PSD of $\mathbf{E}_h$

https://arxiv.org/pdf/2510.13722

    d. **Spectral Residual** (RMSE of log-PSD)
    e. **Effective Resolution** L_eff on E_h spectrum. This is the wavelength where $\frac{E_{pred}}{E_{ref}} < 0.5$

https://arxiv.org/pdf/2606.10642

2. Vertical shear distribution
    a. Calculate many $S_u = \frac{\partial u}{\partial z} \approx \frac{u_2 - u_1}{z_2 - z_1}$ and S_v values and compile into a distribution.
    b. Calculate Wasserstein-1 between the ERA5 and model's distribution (See physMetric thermo stability)

Temporal

1. Temporal power spectrum (persistence)
    a. Fix a single grid point and watch u and v values over time → time series → take 1d of FFT that → power spectrum over temporal frequencies graph
2. Trajectory dispersion
    a. Advect passive agents in 2d through winds.
    b. Compare mean-square displacement vs time and final position spread per altitude

Data Distribution

1. Marginal distribution per pressure level
    a. Get PDFs of u and v per pressure level and compare to ERA5 with wasserstein-1. Plot on log-density axes to inspect tails

        i. Log pdfs of marginals: https://arxiv.org/pdf/2309.15214
        ii. W1 evaluation: https://doi.org/10.1029/2020GL089385
    b. Also report absolute error of extreme quantiles per level. W1 is dominated by the bulk of the distribution, and this is for extreme winds
        i. https://arxiv.org/pdf/2509.26258 (enscale)
2. Conditional distribution per pressure level
    a. Collect historical ERA5 realizations that match a location/time-bucket condition
    b. Fix the condition, generate N samples from N different noise seeds to get distribution of u,v per pressure level
    c. W1 between the two per level and component. Scalar is the average of this value across different conditions

Tiling penalty:
Run suite with single tile evaluation size, and one with multi tile evaluation size, subtract the two to get a tiling penalty.

Baselines

Full Procedure & Calculations

PSD

1. Have vectors $u(x, y)$, $v(x, y)$ at each pressure level. Below is computed per level, and the results can be averaged at the end.
2. Compute $\hat{u}(k)$ and $\hat{v}(k)$, 2D FFT
    a. $\hat{u} = F(u)$,
    b. $\hat{v} = F(v)$
3. Kinetic Energy spectrum value
    a. $E_h(k) = \frac{1}{2}\left(\left|\hat{u}(k)\right|^2 + \left|\hat{v}(k)^2\right|\right)$
        i. Averaged for each $k = \sqrt{k_x^{\,2} + k_y^{\,2}}$
4. Divergence and vorticity spectra
    a. $\hat{\delta}_h(k) = ik_x\hat{u}(k) + ik_y\hat{v}(k)$,
    b. $\hat{\zeta}_h(k) = ik_x\hat{v}(k) - ik_y\hat{u}(k)$
    c. $PSD\left(\delta_h\right)(k) = \left|\hat{\delta}_h(k)\right|^2$
    d. $PSD\left(\zeta_h\right)(k) = \left|\hat{\zeta}_h(k)\right|^2$
5. **Spectral Residual:**
    a. $SR = \sqrt{\frac{1}{N}\sum_k \left(log\, E_{pred}(k) - logE_{ref}(k)\right)^2}$
    b. Gives us SR_δ , SR_E , SR_ζ
6. **Effective Resolution**
    a. $R(k) = \frac{E_{pred}(k)}{E_{ref}(k)}$
    b. L_eff is the wavelength where $R(k) < 0.5$ for 5 consecutive wavenumbers
        i. Tells you up to what resolution the model is good for

Cool: Multi taper/Kaiser window data before FFT